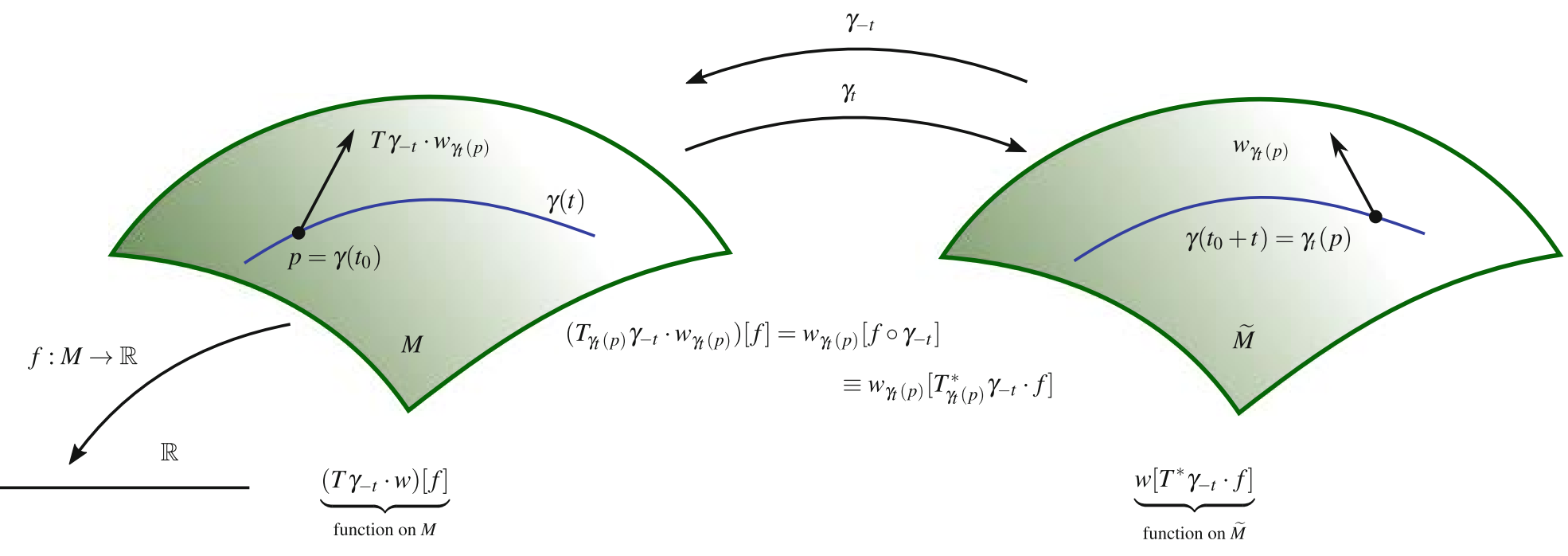
γ is integral curve of vector field v, which gives mappings $\gamma_t : M \to \widetilde{M}$ and $\gamma_{-t} : \widetilde{M} \to M$



$$(T_{\gamma_t(p)}\gamma_{-t} \cdot w_{\gamma_t(p)})[f] = w_{\gamma_t(p)}[f \circ \gamma_{-t}]$$
$$\equiv w_{\gamma_t(p)}[T^*_{\gamma_t(p)}\gamma_{-t} \cdot f]$$

$\underbrace{(T\gamma_{-t} \cdot w)[f]}_{\text{function on } M}$ $\qquad$ $\underbrace{w[T^*\gamma_{-t} \cdot f]}_{\text{function on } \widetilde{M}}$

$$\underbrace{(T\gamma_{-t} \cdot w)[f] \stackrel{?}{=} w[T^*\gamma_{-t} \cdot f])}_{\text{functions on different manifolds}}$$

$$\underbrace{(T\gamma_{-t} \cdot w)[f] = (w[T^*\gamma_{-t} \cdot f]) \circ \gamma_t \equiv T^*\gamma_t \cdot (w[T^*\gamma_{-t} \cdot f])}_{\text{now both functions on } M}$$